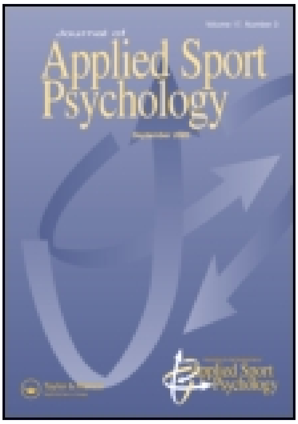

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Perceptual Judgments of Sports Officials are Influenced by their Motor and Visual Experience

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We examined the relation between previous motor and visual experience and current officiating experience of expert judges and referees and their judgments from an embodied cognition viewpoint. A total of 370 sports officials from soccer, handball, ice hockey, and trampoline took part in the study. Analyses revealed that cognitive judgments are related to motor, visual, and officiating experience to different degrees in the analyzed sports. Our findings indicate that, depending on the sport, sports officials should either specialize early in officiating, or gather visuo-motor experience as an athlete or spectator first, and then switch roles to become a sports official.

MOTOR AND VISUAL EXPERIENCE OF JUDGES AND REFEREES

One of the main interests in expertise research on sports judges and referees is to find out more about the factors influencing the quality of their judgments. For instance, a very recent study highlights different aspects that play a determining role in expert performance of referees (Guillén & Feltz, 2011). Although judges observe and assess the quality of performance, and referees are responsible for enforcing the rules (MacMahon & Plessner, 2008), both roles involve officiating a competition or game and, thus, will be collectively referred to as sports officials for the purpose of the present study. Because one single decision can determine the winning team or a gold medal, sports officials face high expectations with regard to making the right decisions. To handle these expectations, they have to reach a high level of expertise. In order to understand how sports officials become experts, it is important to examine factors that enhance and optimize their judgments and decisions.

Extensive studies have focused on external factors influencing the quality of officiating, for instance, the home crowd's noise (Nevill, Balmer, & Williams, 2002), the positioning of the assistant referees (Oudejans, Bakker, Verheijen, Gerrits, Steinbrückner, & Beek, 2005), the positioning of the judges (Plessner & Schallies, 2005), and the judges' own prior decisions (Damisch, Mussweiler, & Plessner, 2006). This knowledge is in contrast to how little is known about internal factors such as motor or visual experience of sports officials. In the present context, internal factors refer to the sensory visual and motor experiences sports officials

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have had themselves, as opposed to experiences using knowledge of rules from memory. If a sports official has been involved in the sport as an athlete or a spectator, thereby gathering a repertoire of unique sensory and motor experiences, will he or she make better decisions? The aim of the present study was to find out if the decision-making performance of expert sports officials from different sports is related to their previous motor and visual experiences and the practical experiences they have had in the role of a sports official on the field or at competitions (officiating experience).

Research has shown that motor as well as visual experience positively influences action perception and, more precisely, the visual analysis of other people's actions (Aglioti, Cesari, Romani, & Urgesi, 2008; Loula, Prasad, Harber, & Shiffrar, 2005). Specifically, in their study, Loula et al. (2005) had participants identify and discriminate point-light depictions of themselves, friends, or strangers performing various actions. Performance was depicted best for self and friend trials and worst for trials of strangers. Because there is a lack of experience in observing strangers, the good performance of judging self and friend trials indicated a positive influence of motor and visual experience on action perception. Given that we do not usually have visual experience of our own movements, motor experience is seen as the influencing factor determining the ability to perceive our own actions. The enhanced ability to judge movements of friends is explained by visual experience.

With reference to judging in figure skating and diving, Allard, Deakin, Parker, and Rodgers (1993) found support that the cognitive tasks of judging are aided by the ability to perform the skills that are meant to be judged. They summarized their results in the following sentence (1993, p. 102): "[. . .] knowing and doing are linked, and doing can facilitate knowing." Similar results were found in a study combining behavioral and neurophysiological data. Participants were required to identify from video footage whether a basketball shot would be a hit or a miss (Aglioti et al., 2008). Despite the fact that both expert players and non-expert players showed a selective increase of motor evoked potentials during observation of basket shots, expert players showed better prediction of movement effects than non-expert players. Although the non-expert players were visual experts (basketball coaches and sport journalists), they were not able to rely on their own movement repertoire for their predictions. The expert basketball players, on the other hand, were able to use their own motor repertoire to judge the effects of the movement kinematics in the videos.

Similar results were found in a study regarding bodily influences on the visual perception of intentions, where motor experience had an impact on the ability to detect a basketball player's intention (Sebanz & Shiffrar, 2009). In a video task, motor experts distinguished fake from true passes better than non-experts. One could argue that the expert basketball players had a more complete repertoire to judge the shots, because it was based on both motor and visual experience. However, other studies have addressed this confounding nature of motor and visual experience by using a blindfold motor learning paradigm (Casile & Giese, 2006; Hecht, Vogt, & Prinz, 2001). Participants, who learned novel and unusual arm/leg movements blindfolded, showed improved visual discrimination of the previously learned movements in a visual discrimination task (Casile & Giese, 2006). The authors also demonstrated that performance differed as a function of skill level. Both discrimination performance of the novel arm/leg movements and the quality of the movement improved during the training phase.

Importantly, visual experience alone has also been shown to improve the quality of perceptual judgments. Using an additional experimental group (visual practice group), Hecht et al. (2001) showed that visual training without motor training also enhances visual judgments. A study in ice hockey and soccer showed that this link between action and cognition can also be transferred to action language (Holt & Beilock, 2006). The task was to judge whether the action described in a sentence matched that of a picture item. All participants (ice hockey and soccer expert and novices) responded most quickly to sentences implying everyday situations, whereas only the sport experts showed the same effect for sport-specific situations.

To sum up, according to an embodied cognition view of perception, action, and decision-making, cognitive processes, including judgments and decision making, are grounded in the body's interactions with its physical context (Wilson, 2002; Wilson & Knoblich, 2005). Researchers investigating the field of embodied cognition have studied how the planning and execution of an action influences concurrent perceptual processes (on-line effects) on the one hand and how motor actions influence perceptual judgments that are temporally separated (offline-effects), on the other hand (Schütz-Bosbach & Prinz, 2007). The above-mentioned studies are considered to focus on offline-effects (temporally separated action and perception processes) of motor action on perceptual judgments. Thus, this bidirectional link between action and perception, with temporally separated processes (offline effects), provides a perspective from which the question, if sports officials benefit from their previous motor and visual experiences, can be explored. In our study, we refer perceptual judgments of sports officials to cognitive aspects and characterize action as the sensory motor and visual experiences sports officials have made in their sport.

Although some studies in the field of embodied cognition have used sport-specific tasks, judgments and decision making of sports officials and a combination of research on judges and research on referees has rarely been studied[1]. Because almost all sport events are guided and decided by sports officials, and therefore depend on the quality of their perception, it appears rather surprising that the ideas and benefits found so far in the field of embodied cognition have barely been investigated in sports judges and referees. Even more surprising, sports officials have scarcely been examined on the basis of motor and visual experience. The fact that the Union of European Football Association (UEFA) introduced and extended the experiment of involving two additional referees into the UEFA Champions League and UEFA Europa League for the 2010/2011 and 2011/2012 seasons to assist in ambiguous foul situations in the penalty area, and so reduce referee errors, shows that there is a need to enhance officiating performance (UEFA, 2009 and 2010).

Research on sports officials has mainly focused on three areas: the demands of officials, the characteristics of decisions, and the different factors influencing decisions and training of elite performers (MacMahon & Plessner, 2008). Studies have shown that officiating experience plays a crucial role in explaining expertise in sports officials. For instance, in a study by MacMahon, Helsen, Starkes, and Weston (2007), soccer referees outperformed athletes in a video-based decision-making task. The best predictor for accuracy in the test was the amount of hours per week in refereeing practice. A study in basketball provided evidence that performance of a certain task is a function of role (Deakin & Allard, 1992). The results showed a superior performance of coaches in a coaching-specific, of referees in a referee-specific, and of athletes in an athlete-specific task. Nevill et al. (2002) further revealed that soccer referees' decisions and their officiating experience show a quadratic relation, suggesting that nonlinear relations should also be taken into account. Based on these results, we would predict officiating experience to be an important factor determining the performance of sports officials in our study, with different types of relations in the analyzed sports.

In the present study we investigate for the first time the internal factors that may relate to the officiating quality of judges and referees in different sports. According to an embodied cognition view, and based on the studies of Aglioti et al. (2008), Casile and Giese (2006), Loula et al. (2005), and Hecht et al. (2001), we predicted a positive relation of previous motor and visual experience to the quality of judgments and decision making of sports officials. Given that the quality of movement performance positively correlates with the accuracy of judging movements (Casile & Giese, 2006), we also hypothesized that greater motor expertise (the league played or trained in as an athlete) correlates with higher quality of the sports officials' judgment. Although some of the described experiments used direct experience (learning experiment) as opposed to previous experience in their study, this study attempts to explore whether their results also hold for previous experience. As it is unclear what the effects of bodily influences might be in the context of perceptual judgments of sports officials, and the results of the above-mentioned studies derived from different movements and sport domains, we considered the possibility that the strength of the internal factors and officiating experience influencing the performance of sports officials may vary, depending on the type of decision task. In a study by Williams and Davids (1995), perceptual strategies and corresponding decision-making processes of experts and non-experts were shown to be task-dependent. Therefore, we predicted that if sports officials are influenced by the type of decision task, the internal factors and officiating experience would determine the performance of sports officials in different ways, depending on the type of sport. Therefore, we conducted a study including four different sports, where the tasks of the sports officials are characterized by a high frequency of rule-based decisions.

We examined soccer referees in an attempt to replicate the study by MacMahon et al. (2007), which found referee experience to positively influence referee performance. Additionally, we predicted motor and visual experience to also positively relate to referee performance. Due to the results found by Nevill et al. (2002), we also considered the possibility of nonlinear relations. To investigate whether our findings generalize across different sports, we further examined handball and ice hockey referees. Characterized by a smaller field and two referees on the field, we hypothesized that handball referee's benefit from officiating as well as motor and visual experience. As ice hockey is a sport that is much faster and requires perceptual judgments to be much quicker, we speculated that motor and visual experience contributes even more to an enhanced ice hockey referee performance. To further test the hypotheses in a different set of judgments, where the judge's role is to evaluate one athlete, as opposed to several on movement quality and not to enforce rules, judges in trampoline were also examined. As studies have shown that motor and visual experience explain enhanced visual sensitivity when observing and discriminating human movement (Loula et al., 2005), we hypothesized especially motor and visual experience to correlate with better trampoline judging performance.

METHOD

Participants

The sports officials were recruited through the referee boards of the respective associations. In total, 370 judges and referees voluntarily took part in a survey. Seventy-seven soccer referees from the top three leagues of Germany participated, with 11.7% of the referees holding an international license (Fédération Internationale de Football Association [FIFA]); 14.3%, 28.6%, and 44.2% had a first-, second- and third-league license, respectively. The 107 team-handball referees were from the first four squads of the Handball League. Of these referees, 9.3% had an international license (International Handball Federation), 18.7% a first-level (referees for men's and women's first-league games), 8.4% a second-level (referees for men's and women's second-league games and some games in the first league to practice), and 63.6% a third-level (referees for men's and women's second-league games and some first-league games for the women) license. In ice hockey, 121 officials of the Ice Hockey Federation took part in the survey, with 14.8% refereeing on an international basis, 23.2% and 26.5% holding a first- and second-league and 35.5% holding a provincial license. In ice hockey, a referee license represents a combination of a certain level as a referee and a certain level as a linesman, indicating that ice hockey referees usually perform both types of officiating. We

Table 1
Means and standard deviations of the internal factors of judges and referees

	Sport							
	Soccer		Handball		Ice hockey		Trampoline	
Variable	***M***	***SD***	***M***	***SD***	***M***	***SD***	***M***	***SD***
Age	28.53	6.22	36.85	7.12	32.45	7.09	31.14	12.18
Officiating experience in years	14.06	6.12	19.51	6.76	11.93	6.80	12.48	9.28
Motor experience in years	10.39	4.63	16.47	7.46	12.02	5.83	14.69	8.38
Visual experience in years	16.6	7.45	16.36	8.28	16.66	8.95	16.22	9.10

had 65 trampoline judges, 56.9% of them held a national license, with 40% of them at the top level and 16.9% at the second level. Another 40% had a provincial license, the data of two judges concerning their license level (3.1%) were missing.

Because experience in the field of sports is referred to as expertise (Ericsson, Charness, & Feltovich, 2006), the determining factors of expertise, number of training years, level and training frequency were taken into account. Therefore, both experience and expertise as a sports official and an athlete were assessed (for a discussion on the relationship between experience and expertise see Williams & Davids, 1995).

Table 1 provides an overview of the sports officials' ages and the officiating, motor, and visual experience they have accumulated over the years. All participants were informed about the purpose of the study and provided informed consent. Their anonymity was protected by coding. The study was approved by the University's Ethical Committee.

Measures

We developed an experience questionnaire to assess the different experiences of sports officials, such as officiating, motor, and visual experience. In this study, officiating experience refers to the practical experience sports officials have made in the role of a judge or a referee. Motor experience is the previous experience the sports officials have made as athletes in the same sport they now officiate in. The sports officials of the current study reported of having been active in their sport as athletes about 10 years ago ($M = 10.21$, $SD = 7.86$). Visual experience refers to watching competitions/games as a spectator as opposed to the visual experience gained in the role of a sports official or an athlete. For an overview of the experience questionnaire see Table 2.

We also assessed the position the sports officials have played (e.g., defense, goal keeper), as we were interested if different positions on the field would show different relations to officiating performance. Furthermore, we tested if motor experience in a different sport but with a similar structure (e.g., soccer and handball) would relate to officiating performance. To examine whether current activity shows a different relation than previous activity, we also asked the participants to indicate if they were still active in the sport as athletes. As all these aspects showed no effects, they were excluded from further analyses. However, we acknowledge that analyses in other than the examined sports may have produced significant results. Altogether, the final experience questionnaire for the current study consisted of three scales with two to three items each.

In addition to the questionnaire, performance measures of the sports officials were collected. Performance measures in soccer, handball, and ice hockey are taken with an observation system

Table 2
Structure and content of the experience questionnaire

Scales	Variables
Scale 1	Officiating experience
	Experience in years (the number of years the sports officials possessed their license)
	License level (level of current license, e. g., A-level)
	Number of games (officiated) per year
Scale 2	Motor experience (as an athlete) in judging/refereeing sport
	Motor experience in years
	Motor experience level (e. g., first league)
	Motor experience training frequency (e.g., three times per week)
Scale 3	Visual experience (as spectator)
	Visual experience (number of competitions/games watched per week/month/year)
	Visual experience in years

in which expert referee coaches regularly watch referees during their games and judge them on different aspects characteristic of good referee performance. These characteristics differ as a function of sport type, but generally performance is rated on the implementation of the game rules (e.g., understanding of the game, foul situations, penalties) and on personal impressions (e.g., referee appearance, positioning, game management). Similar criteria are used both for referees and linesmen. Points are given according to the performance in different categories and then summed up to a final performance measure with higher points representing better performance. The resulting protocols are then used for feedback but also for the determination of referee scheduling for important games. The officiating performance measures are points ranging from 0 to 10 (soccer), 0 to 100 (handball and ice hockey), or 0 to 115 (trampoline), with more points representing better results. The different scales in each sport are due to the different regulations of each sport association. The inter-rater reliability of the handball referee coaches was measured by the intra-class-correlation (ICC) and was found to be good ($ICC = .43, p < .05$), yielding acceptable internal consistency measurements. For trampoline, performance measures were taken from the theory exams. These include questions on rule-based knowledge and the analysis of videotaped trampoline routines, similar to the demands at real competitions. The results of the theoretical tests are used to determine judging performance and, consequently, decide whether a judge can proceed to a higher judging level/license. Data for ICC were not available from the remaining sport associations.

Data Analysis

The data of the experience questionnaires in each sport were coded for analysis by assigning numbers to the different levels of officiating license, motor experience level, and visual experience frequency. After the coding, each item was correlated with the performance measures to get an overview of the data set (see Table 3).

In a second step, a more detailed analysis provided insight into sport-related differences concerning the relation between the performance of sports officials and internal factors. For each of the four sports, standard multivariate linear regression analyses were conducted with eight/nine z-transformed variables (see Table 4) entered equally into the regression model, and with the performance measure as the dependent variable. In order to test for multi-collinearity of the data, the tolerance and VIF (variance inflation factor) values were calculated. The results indicated no severe collinearity problem (O'Brien, 2007).

Table 3
Summary of Correlation Analyses between the Performances of Handball, Soccer and Ice hockey Referees and Trampoline Judges and their Experiences

	Soccer		Handball		Ice hockey		Trampoline	
Variable	*r*	*p*	*r*	*p*	*r*	*p*	*r*	*P*
Officiating experience								
Officiating experience in years	−.25	.030*	.37	.000**	−.03	.760	.13	.351
Sports official license level	−.02	.838	.71	.000**	.28	.002**	.06	.643
Number of competitions/games per year (referee)	−.11	.353	.35	.000**	.00	.985	−.07	.593
Number of competitions/games per year (linesman)					.19	.034*		
Motor experience (as an athlete) in officiating sport								
Motor experience in years	−.15	.185	−.19	.052	.19	.038*	.28	.027*
Motor experience level	−.05	.683	−.01	.931	.07	.439	.18	.160
Motor experience training frequency	−.05	.649	.26	.007**	.12	.189	.21	.102
Visual experience (as spectator)								
Visual experience (watching frequency)	.25	.026*	.09	.359	.05	.569	−.18	.159
Visual experience in years	.05	.690	.21	.029*	.00	.983	.16	.221

*$p < .05$. **$p < .01$.

Table 4
Summary of Multiple Regression Analyses predicting Performance of Handball, Soccer and Ice hockey Referees and Trampoline Judges

	Soccer		Handball		Ice hockey		Trampoline	
Predictor	***B***	***β***	***B***	***β***	***B***	***β***	***B***	***β***
Officiating experience								
Officiating experience in years	−.05	−.49**	.09	.18*	.02	.10	.07	.07
Sports official license level	.11	.19	1.91	.60**	.09	.27**	.45	.05
Number of competitions/games per year (referee)	−.06	−.18	.03	.02	.12	.33*	−1.06	−.11
Number of competitions/games per year (linesman)					.29	.58**		
Motor experience (as an athlete) in officiating sport								
Motor experience in years	−.01	−.08	−.05	−.12	.05	.23*	.19	.18
Motor experience level	.05	.09	−.03	−.02	−.06	−.11	.34	.06
Motor experience training frequency	−.05	−.14	.18	.11	.05	.11	.41	.11
Visual experience (as spectator)								
Visual experience (watching frequency)	.31	.34**	−.27	−.05	.22	.12	−1.95	−.12
Visual experience in years	.02	.19	.03	.07	.01	.06	−.03	−.03

$^*p < .05$. $^{**}p < .01$.

To determine whether officiating performance shows a nonlinear relation to officiating, motor, or visual experience in years, quadratic and cubic regression analyses were conducted in each sport and for each variable separately (see Table 5). Therefore, in each regression analysis officiating performance served as a dependent variable and one experience variable (e. g. motor experience) served as a predictor variable. Statistical significance was set at p < .05 for all statistical tests.

RESULTS

We asked whether internal factors, such as motor and visual experience and officiating experience, relate to sports officials' performance. It was hypothesized that greater officiating, motor, and visual experience leads to a better performance of the sports officials. The resulting

Table 5
Summary of Linear, Quadratic, and Cubic Regression Analysis for Variables Predicting Officiating Performance in Each Sport

	Soccer	Handball	Ice hockey	Trampoline
Officiating experience in years	Cubic $R^2 = .22$ $p < .01$	Cubic $R^2 = .17$ $p < .01$	No relation	No relation
Motor experience in years	No relation	Linear $R^2 = .04$ $p = .05$	Linear $R^2 = .04$ $p < .01$	Linear $R^2 = .08$ $p < .05$
Visual experience in years	Cubic $R^2 = .24$ $p < .01$	Cubic $R^2 = .13$ $p < .01$	No relation	No relation

multiple regression models significantly predict the performance of soccer ($R^2 = .24$; $df = 8$, $N = 77$; $p = .014$), handball ($R^2 = .55$; $df = 8$, $N = 94$; $p < .001$), and ice hockey referees ($R^2 = .24$; $df = 9$, $N = 119$; $p < .001$), but not of trampoline judges ($R^2 = .12$; $df = 8$; $N = 65$; $p = .694$). The regression coefficients (B) are shown in detail in Table 4.

Table 4 shows that officiating experience relates to officiating performance in handball and ice hockey, either concerning the number of years as a sports official (handball) or the license level and the number of games refereed (ice hockey). In soccer, officiating experience is negatively related to officiating performance. However, this result is probably due to the cubic relationship, which will be addressed later. Motor experience (in years) significantly predicted ice hockey referee performance and showed a positive significant correlation with trampoline judging performance. In soccer, a high amount of visual experience was related to high referee performance quality.

To assess the predictive ability of the regression models, we performed a cross-validation. Specifically, we conducted a regression analysis with all even-numbered participants. After fixing these values, we generated predictions for the odd-numbered participants. Using this cross-validation procedure the model fit for the even-numbered and odd-numbered participants for soccer referees was $R^2 = .64$ and $R^2 = .42$, for handball referees, $R^2 = .62$ and $R^2 = .56$, for ice hockey referees, $R^2 = .21$ and $R^2 = .36$, and for trampoline judges, $R^2 = .36$ and $R^2 = .39$, respectively. The results show that the regression models predict officiating performance quite well. A path analysis, controlling for collinearity and testing also direct/indirect effects with performance measure as the dependent variable, revealed similar results.

The results of the nonlinear regression analyses for each sport and each experience factor reveal that there is a cubic effect for officiating and visual experience in soccer and handball (see Table 5). In soccer, officiating performance increases with a minimum of 10 years of officiating experience. Furthermore, officiating performance increases with a greater amount of visual experience until about 11 years of visual experience are reached and for visual experience of more than 30 years. But between 11 and 25–30 years of visual experience there is a small decrease in performance. In handball, officiating performance continuously increases with a greater amount of officiating experience. The steepest gradient, as so the greatest increase in officiating performance, can be observed until about 11 years and after 25 years of officiating experience. Furthermore, an increasing amount of visual experience, until about a peak of 23 years, is related to higher officiating performance; from then on the curve shows a decrease in performance.

Within handball, ice hockey, and trampolining, motor experience and officiating performance can be described best by a linear relationship (see Figure 1). Whereas in soccer there is no relation between motor experience in years and officiating performance, in handball there is a negative and in ice hockey and trampoline a positive linear relation.

DISCUSSION

The aim of the current study was to examine the relation between previous motor and visual experience and current officiating experience of expert sports officials and their judgments from an embodied cognition viewpoint. Besides officiating experience, we additionally predicted motor and visual experience to positively relate to officiating performance. Sports officials from three team sports, soccer, handball, and ice hockey and one technical sport, trampoline, were examined by relating their performance as officials to their practical experiences as judges or referees, to their motor experiences as athletes, and to their visual experiences as spectators.

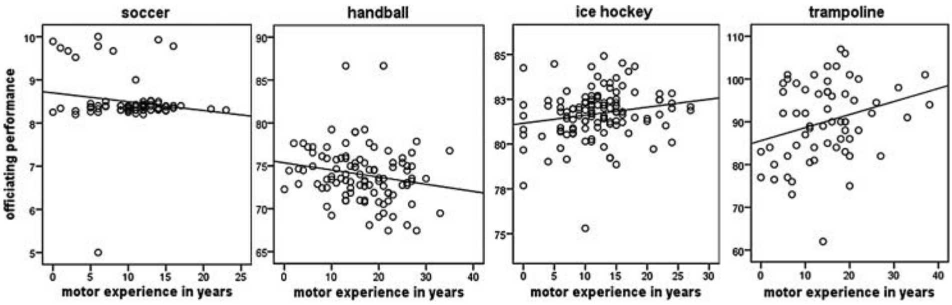

Figure 1. Relation of motor experience in years and officiating performance (points) compared between soccer, handball, ice hockey, and trampoline.

The hypothesis that officiating, motor, and visual experience positively relate to the performance of sports officials was partly confirmed. Depending on the sport, officiating, motor, and visual experiences were differentially important factors relating to the performance of sports officials.

Officiating Experience

According to the data of the sport-specific linear regression analyses, handball and ice hockey referees should consider an early specialization as a sports official to ensure many years of practical experience at an early age and proceed to a higher license level. Hence, referee experts in handball and ice hockey can be characterized by a higher license level. Furthermore, our results show that the performance measures obtained from the sports associations are coherent with the officiating experience concerning the license level. Interestingly, nonlinear (cubic) regression analyses additionally revealed that in soccer a minimum of 10 years of officiating experience should be reached to enhance officiating performance. This non-linear relation was also found in the study by Nevill et al. (2002) on soccer referees. Furthermore, a high game-refereeing frequency should be reached in ice hockey to increase practical experience. This confirms and extends the study by MacMahon et al. (2007) and furthermore, underlines that the educational referee systems of handball and ice hockey are effective in their role of training their judges and referees.

Motor Experience

As motor experience (in years) significantly predicted ice hockey referee performance, learning how to play ice hockey and experiencing the characteristics of an ice hockey game as an athlete, could be useful indications for good officiating performance. The fact that ice hockey is a much faster game than the others, requiring perceptual judgments to be much quicker, seems to ask for experience as athletes to meet these difficult tasks as sports officials. Similar implications are suggested by the positive relations between trampoline judging performance and motor experience in trampolining. Judging in trampoline, based on detecting and discriminating different skills, seems to relate to experiences incorporating the feeling of biomechanical characteristics and physical forces as a trampolinist. Furthermore, in handball, ice hockey, and trampoline a linear trend was found for the relationship between motor experience in years and officiating performance. The data of ice hockey referees and trampoline judges indicated that a higher amount of motor experience in years as an athlete is associated with better officiating performance. Although this positive trend was not found for soccer referees in this study, it was reported by MacMahon and Plessner (2008), from a study with referees, that referees felt that their experience gained as athletes helped them as referees to know how a player thinks, to interpret player intentions, and to spot deceptions. In contrast to our hypothesis, motor experience and officiating experience showed a negative relation for handball referees. Further studies are needed to examine this finding in more detail.

Visual Experience

Visual experience was considered even more important in soccer than in the other sports to achieve high referee performance quality. Again, visual experience in years and officiating performance was best described by a cubic relationship, both in soccer and in handball. According to this cubic relationship in soccer, officiating performance is barely enhanced between 11 and 25–30 years of visual experience. Before reaching 11 years and after 25–30 years of visual experience, officiating performance shows a high increase.

Surprisingly, although visual experience was hypothesized to predict officiating experience in all four sports, it significantly predicted the performance of soccer referees only. This could be due to the possible confounding character of visual experience, as officiating or performing the sport as an athlete is usually automatically accompanied by visual experience. Even though visual experience in this study (watching games or competitions without having to judge them) was separated from visual experience attained during officiating or practicing the sport as an athlete, these factors may still be intrinsically linked in their contributions to implementing officiating judgments.

These findings on motor and visual experience partly extend earlier studies by Loula et al. (2005) to the domain of sports officiating. This is one of the first studies showing that there is a relationship between the motor and visual experiences of expert sports officials and their decision-making performance. Therefore, especially ice hockey referees and trampoline judges should gather experience as athletes over many years before becoming judges or referees, in order to be able to show higher quality judgments and decisions. Poplu, Baratgin, Mavromatis, and Ripoll (2003) suggest that motor practice develops perceptual and conceptual knowledge, which then enhances decisions and reaction times. Several studies have revealed that motor experts show a special perceptual sensitivity to actions (see special issue in the *Journal of Sports Sciences* on observational learning by Hodges and Williams, 2007). If further replicated, sport associations could take these results into account and develop criteria accordingly for the entry into the educational programs of sports officials. For instance, to be selected for the program as an aspiring sports official, sport associations could require previous motor experience (a certain amount in years or/and a certain level), as it is currently conducted in judo and dressage in horse riding in some countries. Additionally, certain situations that are frequently debated (e.g., the dive in soccer) could be systematically trained by the sports officials in the role of an athlete and an observer, to facilitate the progress of putting themselves into the specific situation.

Officiating experience was shown to predict handball and ice hockey referee performance. The fact that motor experience also predicted and positively correlated with their officiating performance could explain the relatively high age of these expert referees performing at a high level (see Table 1). The practical implication of this result is that in some sports officials should gather experiences as athletes first and then specialize in officiating, where they will need to collect many years of experience and proceed to higher license levels before becoming expert sports officials. These findings are supported by work from MacMahon, Starkes, and Deakin (2009), where role transitions (e.g., players who become referees) were able to explain performance in decision-making tasks better than role-based differences, contradicting the results found by Deakin and Allard (1992).

When comparing dependent and independent measures of the different sports, there are some methodological issues that have to be taken into account. First, the performance measures of the sports officials were provided by the respective sport associations. With the exception of handball, no further validation of the performance measures was possible, due to limited access to the data. Further studies should raise the possibility to collect performance measures in the laboratory, for example, by conducting video tests. Furthermore, the evaluation process of the sports officials differs slightly depending on the sport (referee observation system versus theoretical exams based on video analysis). Although, as explained in the data-analysis, judges and referees share the same tasks—performing cognitive judgments based on visual movement analysis—and data was z-transformed, these differing evaluation processes could explain the partly inconsistent results found in this study in the detailed analysis. However, this study was meant to obtain a first overview of this topic and explore relations. In the current study, we set our focus on the overall performance of a sports official. Nevertheless, other basic areas of performance have also been discussed, showing the complexity of a sports officials' task (for an overview, see Mascarenhas, Collins, & Mortimer, 2005). Second, independent measures, such as license and motor experience level, refer to different skill levels in the different sports. In this case, z-transformed data provided a good tool to overcome this problem and permit a comparison across sports.

To the best of our knowledge, our study is the first to provide an overview of internal factors relating to the performance of sports officials of different sport domains from an embodied cognition viewpoint. However, as the current article is based on correlation studies, in a next step, the results should be replicated experimentally. The question of interest is if sports officials with a higher amount of officiating, motor, or visual experience show better performance in a sport-specific decision-making task. In a subsequent step, experiments should be conducted to find out whether sports officials can enhance their judgments and decisions by learning the motor task themselves, by frequently observing other judges or referees, or even by frequently observing other athletes practicing the movements to be judged. Future studies could enhance the experimental design by differentiating more between visual and motor experience and completely occluding one or the other (e.g., Casile & Giese, 2006) or combining them to form one variable. The idea of examining judges and referees in one study, a rather new approach in this field, shows that there is potential for interesting and innovative research. With its overview of judges and referees of different sports, this study provides a basis from which more detailed experiments could be conducted. Focus could be laid especially on the different effects of the sports officials' previous experiences when refereeing sport games as opposed to judging in technical sports.

Research on judgment and decision making of sports officials should take into account that perceptual judgments cannot be separated from sensorimotor experiences. When addressing the goal of enhancing perceptual judgments of sports officials, the possibility should be considered that they use their own body to support perception and cognition, and therefore, their perceptual judgments. To get a complete picture of how sports officials judge and decide in the field of sport, long-term research should combine different theories of psychology, economics and embodied cognition. For instance, interaction between different internal factors, such as knowledge, memory, and experience, could be examined. Furthermore, it could be interesting to take a look at internal factors interacting with external factors. For example, does the influence of crowd noise differ as a function of motor and visual experience?

In summary, there are several internal factors relating to the performance of sports officials. Depending on the sport, the internal factors seem to contribute differently to officiating performance, confirming the hypothesis that embodied cognition in judges and referees is task dependent. Practically speaking and if experimentally replicated, soccer referees' performance may be enhanced by watching a great amount of games over many years and handball referees might benefit from deciding early to become a referee. At the same time the performance of ice hockey referees, just as that of trampoline judges could be enhanced by also starting their career as athletes quite early, as winning medals as athletes could lead to better performance as referees and judges.

FOOTNOTE

1. One study was conducted to examine the influence of the observed person's physical aspects (height) on perceptual judgments of referees (see van Quaquebeke & Giessner, 2010) rather than motor experience per se.

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